Proof: Properties (b)–(e) are considered in the exercises.

- Let B and C be $n \times o$ and $o \times p$ matrices respectively.
- Let $A = [a_1 \; a_2 \; \cdots \; a_n]$ $B = [b_1 \; b_2 \; \cdots \; b_o]$ $C = [c_1 \; c_2 \; \cdots \; c_p]$

$BC = [Bc_1 \; Bc_2 \; \cdots \; Bc_p]$ (By defn. of matrix multiplication)

$A(BC) = A[Bc_1 \; Bc_2 \; \cdots \; Bc_p] = [A(Bc_1) \; A(Bc_2) \; \cdots \; A(Bc_p)]$ (By defn. of matrix multiplication) $= [(AB)c_1 \; (AB)c_2 \; \cdots \; (AB)c_p]$ (By validity of eqn. 1) $= AB[c_1 \; c_2 \; \cdots \; c_p] = (AB)C$ (By defn. of matrix multiplication) □

$$\boxed{\therefore A(BC) = (AB)C}$$

- The left-to-right order in products is critical because AB and BA are usually not the same.
- The columns of AB are linear combinations of the columns of A, whereas the columns of BA are linear combinations of the columns of B.
- If $AB = BA$, we say that A and B **commute** with one another.

Example 7: Let $A = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 4 & 3 \end{bmatrix}$. Show that these matrices do not commute. That is, verify that $AB \neq BA$.

Solution: Let $B = [b_1 \; b_2]$. By defn. of matrix multiplication,

$AB = [Ab_1 \; Ab_2]$

$Ab_1 = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 2 \\ 4 \end{bmatrix} = 2\begin{bmatrix} 5 \\ 3 \end{bmatrix} + 4\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ (By defn. of Ax)

$= \begin{bmatrix} 10 \\ 6 \end{bmatrix} + \begin{bmatrix} 4 \\ -8 \end{bmatrix}$ (Scalar multiplication)

$= \begin{bmatrix} 14 \\ -2 \end{bmatrix}$

$Ab_2 = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \cdot 0 + 1 \cdot 3 \\ 3 \cdot 0 + (-2) \cdot 3 \end{bmatrix}$ (By Row vector product)

$= \begin{bmatrix} 3 \\ -6 \end{bmatrix}$